

Inborn Errors of Immunity - Factoid Sheet

Week 8 • Core Lecture 2 • current Dolen deck controls

Algorithm: infection pattern → phenotype → screening labs → targeted second-line testing → genotyping / specialist input.

Warning Signs + First-Line Screen

Child red flags
Family history • abnormal TREC • ≥4 ear/yr • ≥2 serious sinus/yr • ≥2 pneumonias/yr • deep abscess/sepsis • persistent thrush/fungal skin disease • ≥2 months antibiotics with poor response • failure to thrive • IV antibiotics.

First line
CBC + diff • quantitative Ig levels **GAME** • maybe delayed-hypersensitivity skin testing • HIV-1 testing • C3/C4/CH50/AH50 • review newborn TREC.

Screen	What to extract
CBC/diff	Absolute lymphocyte + neutrophil counts; platelet number + size; smear. Low ALC → SCID concern. Microthrombocytopenia → WAS clue.
GAME	All near-zero → SCID; ≥2 low classes/no explanation → “CVID” phenotype; high/normal IgM + other classes very low → hyper-IgM.
TREC	Congenital lymphopenia screen, not SCID-specific. SCID, DiGeorge, A-T, syndromes, and secondary causes can lower it.
Complement	CH50 classical; AH50 alternative; add C3/C4. C1 inhibitor quantity/function for angioedema; C1q for acquired angioedema.

Second / Third Line

Second line	High-yield use
Antibody function	Isohemagglutinins; vaccine/common-infection IgG titers
Flow cytometry	T/B/NK counts; SCID phenotype; CD40L; selected neutrophil studies
Mitogen / antigen stimulation	In-vitro lymphocyte proliferation
DHR	Oxidative burst; CGD shows essentially no stimulated shift; X-linked carrier can show 2 populations
Third line	Individual genes/panels; exome/genome; specialist-directed testing

Phenotype Map

Phenotype	Pattern	Examples / anchors
Combined / SCID	Viral + bacterial + fungal + protozoal; low ALC	SCID; A-T
T-cell dysfunction	Viral/fungal/protozoal	DiGeorge; WAS; CMC/APS-1; UNC93B/TLR3 mimic
Hyper-IgM	IgM nl/high; other classes ~0	CD40L deficiency high-yield; NEMO mimic
Antibody deficiency	Mostly bacterial; opsonization problem	XLA; IgA deficiency; “CVID”; THI
Complement	Bacterial / immune-complex / angioedema patterns	Terminal pathway → Neisseria
Phagocyte	Deep abscesses, pneumonia, osteomyelitis	Aspergillus, Nocardia, Serratia, Staph; CGD/LAD/CHS

SCID + Named IEI Recognition

SCID Immunophenotypes - Use the Current Slide 36 Table

Phenotype	Deck genes	Why
T+ B+ NK+	p56Lck	Severe signaling defect can exist despite cell presence
T- B+ NK-	IL2RG, JAK3	B cells do not require these pathways
T- B+ NK+	IL7RA; CD3δ/ε/γ	T-lineage development/signaling selectively disrupted
T- B- NK-	ADA	Metabolic defect injures lymphocyte lineages broadly
T- B- NK+	RAG1/2; Artemis	T/B receptors need V(D)J; NK cells do not

Omenn: phenotype can be highly variable, even T+B-NK+. Do not force it into one row.

Named IEIs - Lecture Anchors

IEI	Lecture cue
“CVID”	Working phenotype: ≥2 low Ig classes + no alternate explanation. BAFF/TACI/APRIL high-yield. Genotype when possible. Gamma-globulin replacement; APDS may be targeted with leniolisib.
XLA	Antibody-deficiency phenotype; very low Ig; BTK is model targeted gene test. Companion: absent B cells / boy after maternal IgG wanes.
Selective IgA deficiency	Antibody-deficiency phenotype. Companion adds common/asymptomatic + transfusion-reaction clue.
DiGeorge	T-cell-dysfunction phenotype; abnormal TREC / congenital lymphopenia differential.
WAS	Microthrombocytopenia on CBC; T-cell-dysfunction phenotype. Companion: eczema + recurrent infections.
CGD	DHR failure; phagocyte phenotype; Aspergillus/Nocardia/Serratia/Staph; hypergammaglobulinemia possible.
LAD	Phagocyte phenotype. Delayed cord separation can occur but faculty warns not to anchor on LAD-I alone.
Terminal complement deficiency	Recurrent Neisseria .
Hereditary/acquired angioedema	C1 inhibitor quantity/function; C1q helps acquired angioedema workup.
Hyper-IgM	IgM normal/high, other classes almost zero; CD40L deficiency high-yield.
THI / Chédiak-Higashi	Named in phenotype/objective lists; no full dedicated current-deck disease slide here.

Stealth Objective

Recognize, don't over-memorize: ataxia-telangiectasia • MSMD • hyper-IgE syndrome(s) • CMC / APS-1 • IRAK4/MyD88 deficiency • UNC93B/TLR3 deficiency.

Exam logic: broad opportunists + low ALC = combined • bacterial = antibody (but check complement/phagocyte mimics) • Neisseria = terminal complement • deep catalase-type abscess pattern = phagocyte • T/B/NK pattern = derive from development.

Primary: current 54-slide Dolen deck. Companion-only classic disease details are labeled as secondary enrichment.